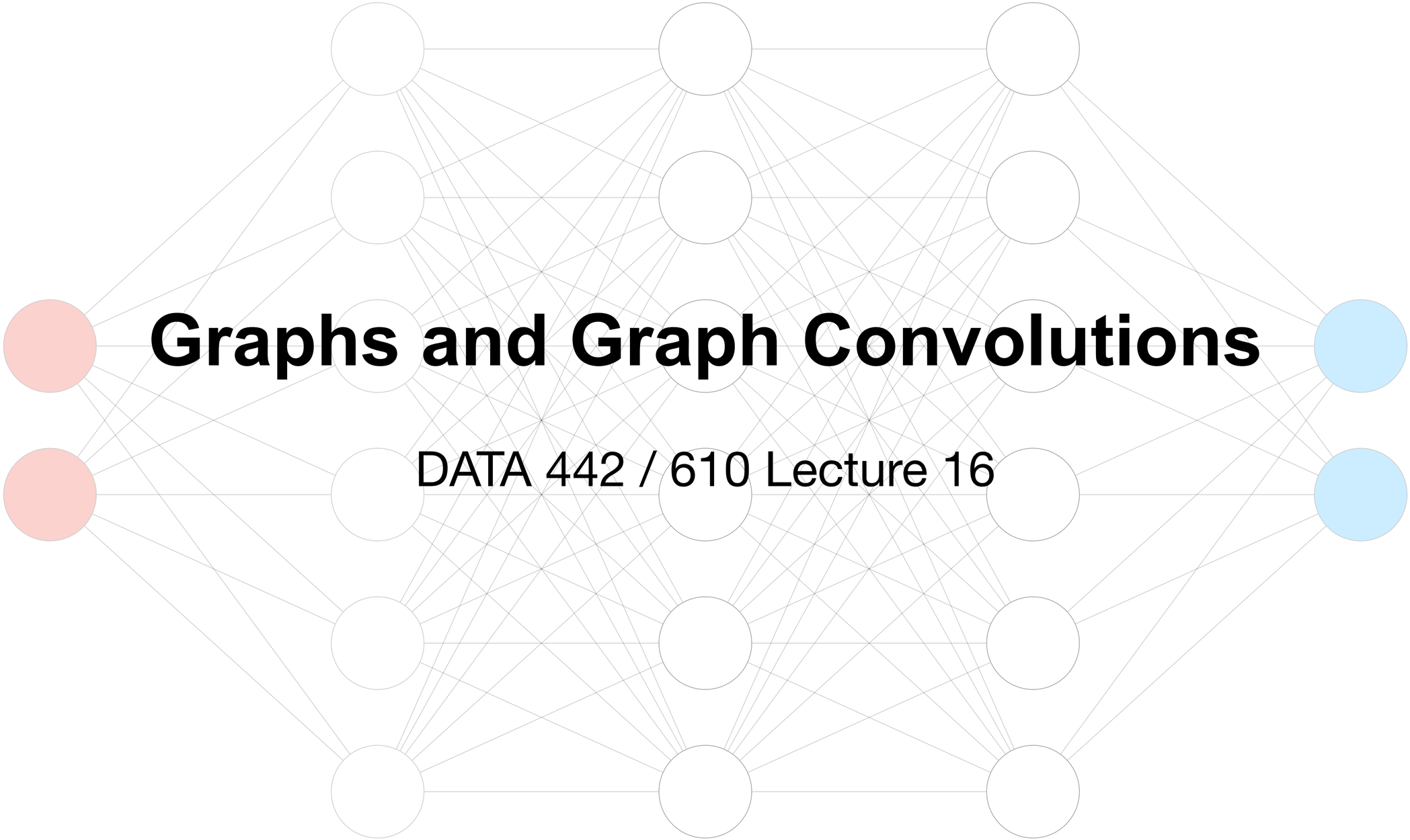


Graphs and Graph Convolutions

DATA 442 / 610 Lecture 16

Outline

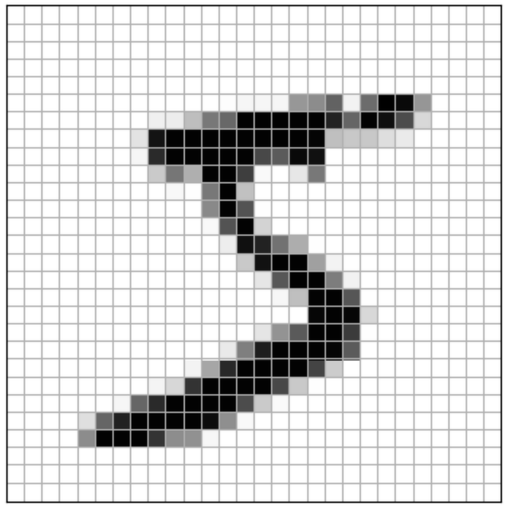
- Graphs
- Graph Convolutions
- Demo with networkX

Relevant reading: ArXiv:2104.13478

Data types

Image data





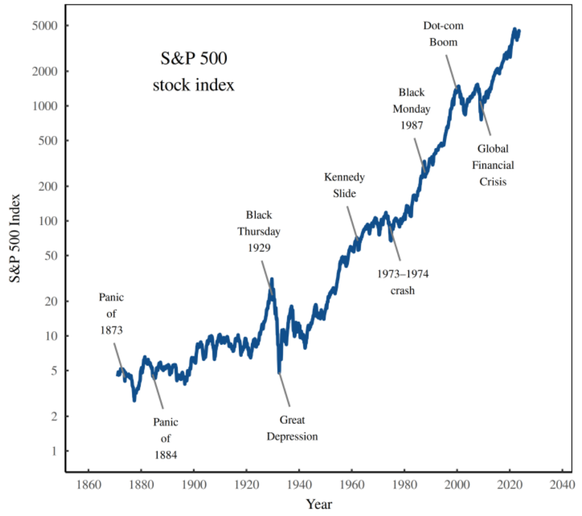
Data is represented as an array of pixels

CNN learns kernels that identify features from local patches

Sequential data

$$\left\{x^{(1)}, x^{(2)}, x^{(3)}, \ldots, x^{(t)}\right\} \quad t > 1$$

any text or sentence data that we could possibly imagine

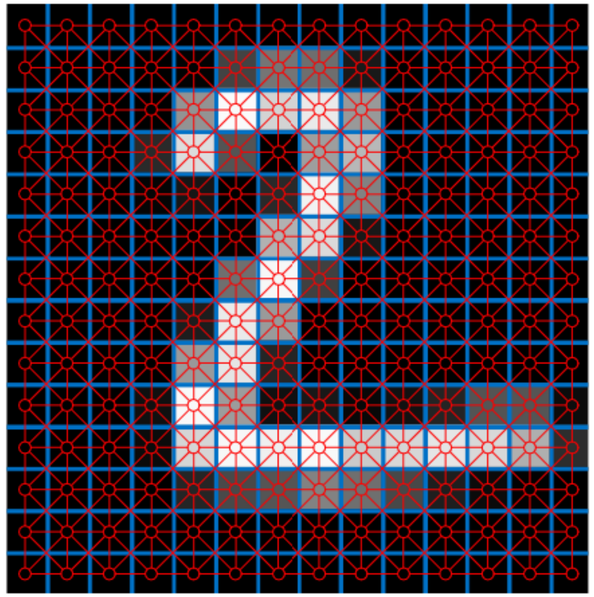


Data is represented as a sequence of feature vectors

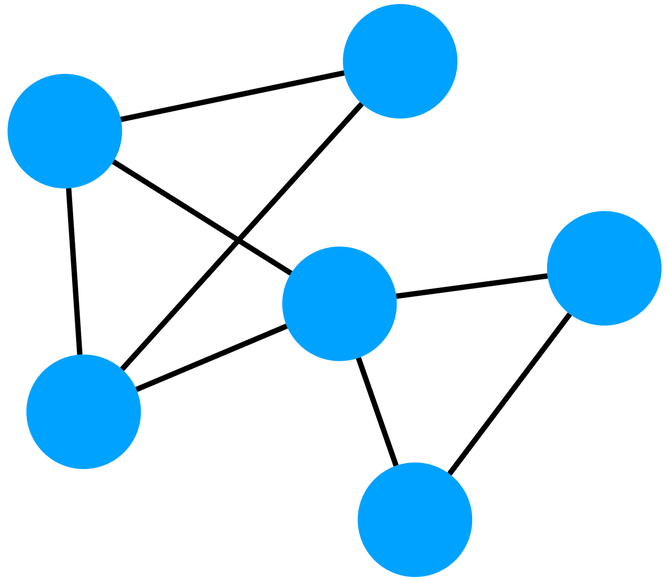
RNN learns functions that extract and recall info from sequence elements

Data Types

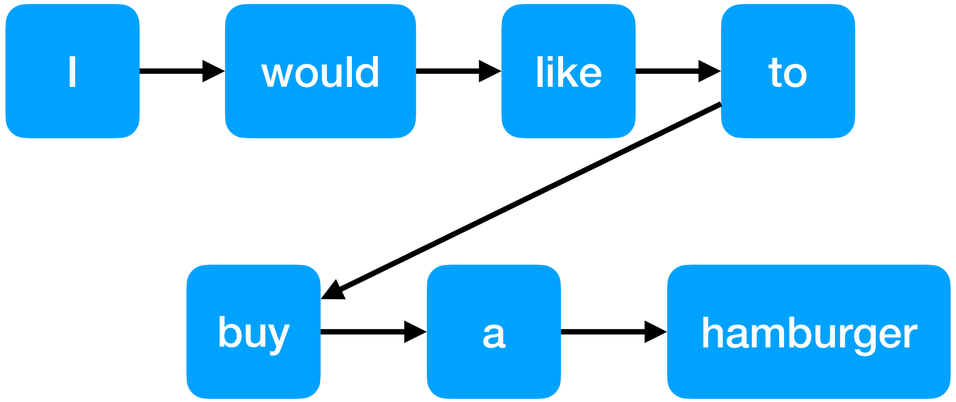
Image data



Graph data

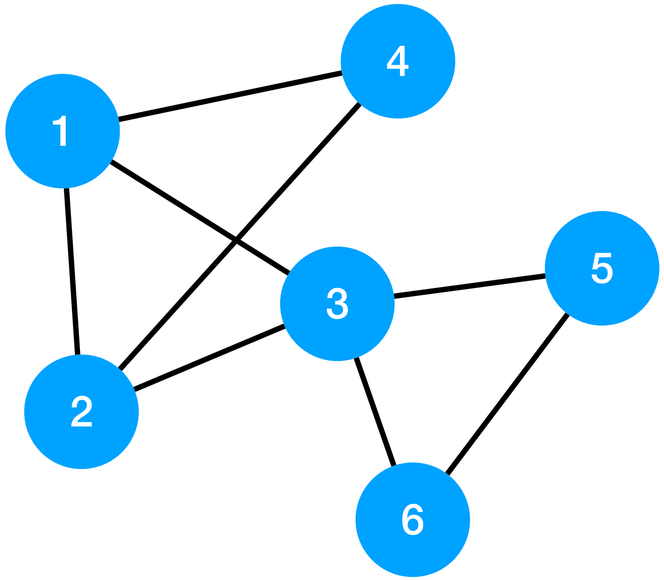


Sequential data



A **graph** is a structure containing **nodes** V and **edges** $E \subseteq V \times V$

Graphs



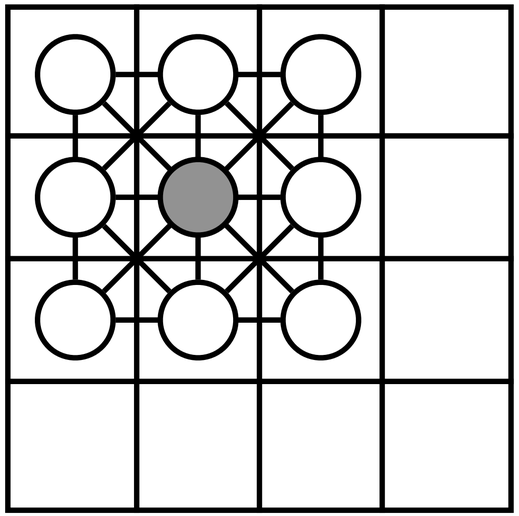
Nodes: $V = \{1,2,3,4,5,6\}$

Edges: $E = \{(1,2), (1,3), (1,4), (2,3), (2,4), (3,5), (3,6), (5,6)\}$

Adjacency matrix: $A_{ij} = \begin{cases} 1 & \text{if edge between } i \text{ and } j \\ 0 & \text{otherwise} \end{cases}$

$$A = \begin{pmatrix} 0 & 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 \end{pmatrix}$$

Other data types as graphs

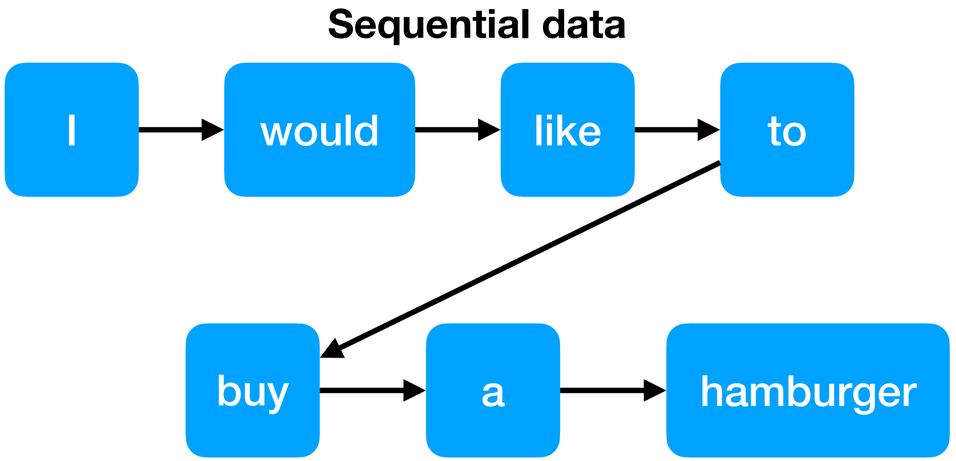


$$V = \{11, 12, 13, 14,\ 21, 22, 23, 24,\ 31, 32, 33, 34,\ 41, 42, 43, 44\}$$

$$E = \{(11,12), (11,21), (11,22), (12,13), (12,21), (12,22), (12,23), \ldots\}$$

$$A = \begin{pmatrix}
0 & 1 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
1 & 0 & 1 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 1 & 0 & 1 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
1 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
1 & 1 & 1 & 0 & 1 & 0 & 1 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 1 & 1 & 1 & 0 & 1 & 0 & 1 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 1 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 1 & 1 & 0 & 1 & 0 & 1 & 0 & 1 & 1 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 0 & 1 & 0 & 1 & 0 & 1 & 1 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 0 & 1 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 0 & 1 & 0 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 1 & 0
\end{pmatrix}$$

Other data types as graphs



$$A = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

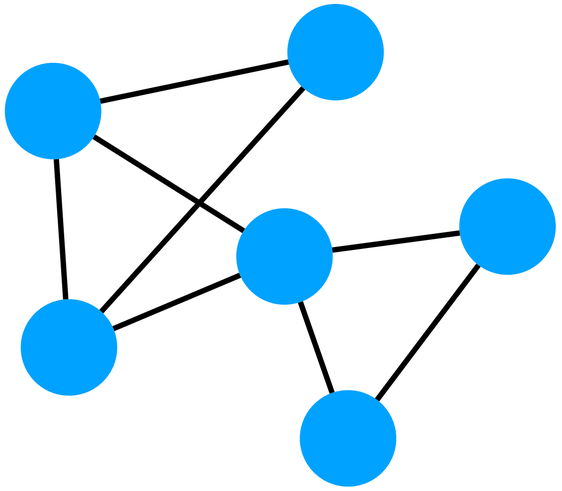
$$V = \{\text{I}, \text{would}, \text{like}, \text{to}, \text{buy}, \text{a}, \text{hamburger}\}$$

$$E = \{(\text{I}, \text{would}), (\text{would}, \text{like}), (\text{like}, \text{to}), \ldots\}$$

Sequential data may be a **directed graph** due to time-ordering

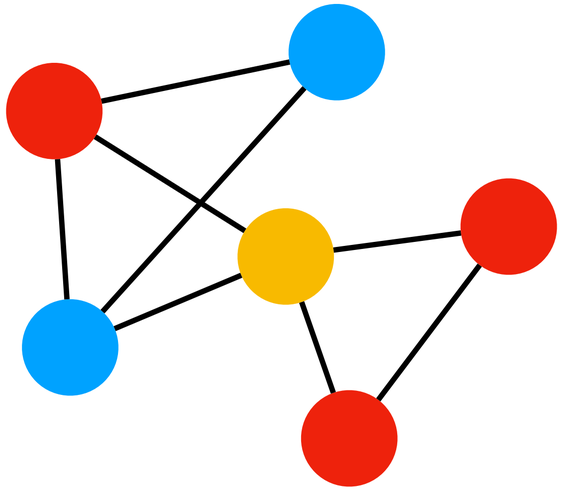
$$A_{ij} \neq A_{ji}$$

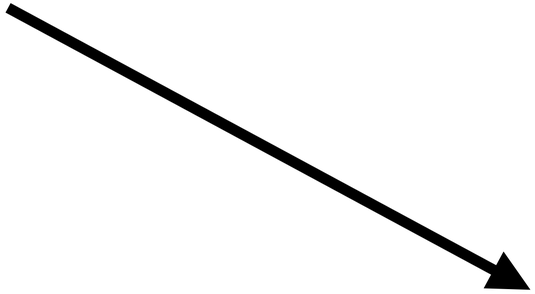
Types of graph problems





Node labeling



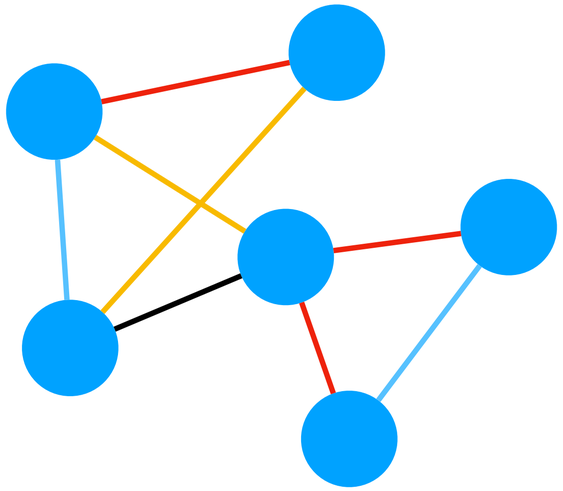




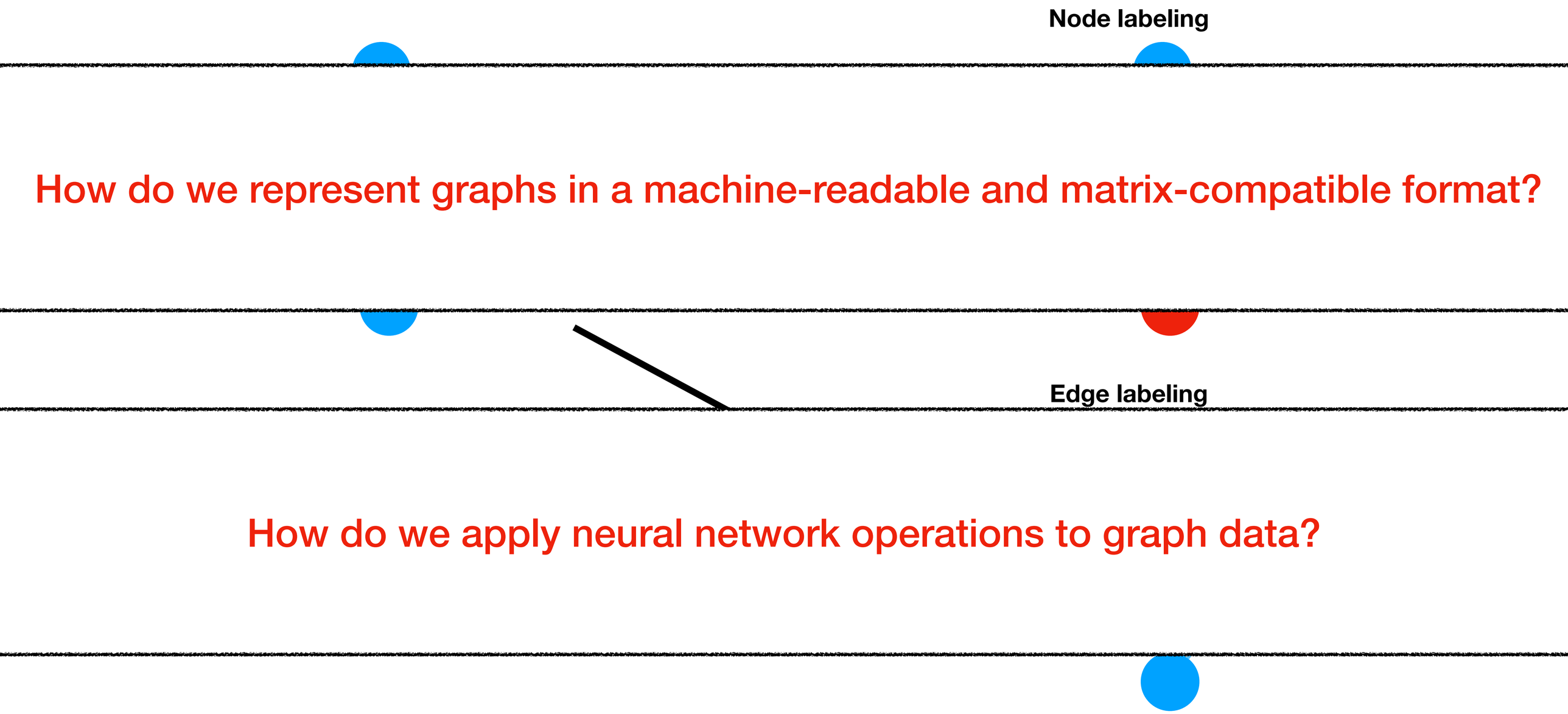
Graph-level labeling

“This is a car”

Edge labeling



Types of graph problems



Let’s go talk about graphs



That’s all!

- Slides will be posted on the course website: https://colen-teaching.github.io/DATA442-Fall-2026
- Lab demos will also be posted
- ***PS3*** due November 8
- **REMEMBER:** TA Office Hours Tuesday 12-1pm, Student Success Center